Notes: Blickle, Brunnermeier & Luck (RFS, 2024)

Who Can Tell Which Banks Will Fail?

Contents

1 Big picture

- **Question.** In a systemwide bank run, can depositors distinguish banks that will fail from banks that will survive? Are banks better informed than regular depositors?
- **Historical setting.** The paper studies the German banking crisis of 1931, one of the major financial crises of the Great Depression.
- **Why this setting is powerful.** Germany had no deposit insurance, no modern capital or liquidity regulation, and limited government intervention before the bank holiday. Depositors had strong incentives to withdraw if they expected losses.
- **New data.** The authors digitize monthly bank balance sheets from the *Deutscher Staats- und Preussischer Reichsanzeiger*, covering more than 120 banks and over 50% of German banking-sector assets.
- **Main fact.** Deposits fall by roughly 20% during the run, but total deposit outflows are about the same for failing and surviving banks.
- **Key distinction.** Regular deposits—retail and nonfinancial wholesale deposits—do not differentiate failing from surviving banks. Interbank deposits do.
- **Main result.** Banks withdraw interbank deposits almost exclusively from banks that later fail. Failing banks are effectively shut out of the interbank market.
- **Prediction.** Interbank deposit flows predict future bank failure. A simple model using interbank outflows reaches AUC values above 0.8.
- **Interbank-market mechanism.** The interbank market does not freeze for all banks. Surviving banks with deposit outflows continue to borrow from banks with deposit inflows.
- **Interpretation.** Banks appear to have superior information about which other banks are weak, and they use that information while still providing liquidity to viable banks.
- **Economic takeaway.** During a systemic run, uninformed depositors may run broadly, while informed intermediaries selectively discipline weak banks. This means interbank markets can simultaneously transmit information and provide liquidity.

2 Data and measurement

2.1 Bank-level balance sheet data

- The baseline sample covers an average of more than 120 banks per month in early 1931.
- Included banks are Berlin banks, regional credit banks, Girozentralen, and Landesbanken.
- The data contain more than 70 balance sheet items at monthly frequency.
- The sample covers over half of total banking-sector assets and over three quarters of commercial and industrial lending.
- The sample does not include local savings banks, mortgage banks, or private investment banks/brokers.

Interpretation: The data are unusually granular for a systemwide historical run. They allow the authors to study depositor behavior across many banks exposed to the same macro shock.

2.2 Types of deposits

The paper separates liabilities into categories that are central for the identification of informed withdrawals:

- **Regular deposits:** deposits from retail and nonfinancial wholesale depositors.
- **Domestic interbank deposits:** deposits from other German banks and credit institutions.
- **Demand deposits:** regular and interbank deposits with no maturity or maturity below seven days.
- **Time deposits:** regular and interbank deposits with maturity above seven days.
- **Foreign-currency deposits:** approximated from confidential Reichsbank filings from 1929–1930.

Interpretation: The core empirical comparison is between regular depositors and banks as depositors. If anyone can identify fragile banks, the paper asks whether it is regular depositors, interbank lenders, or the stock market.

2.3 Assets and liquidity

The asset side is separated into:

- illiquid loans and covered bonds;
- high-quality liquid assets, including cash, central bank reserves, and Reichsbank-eligible securities;
- lower-quality liquid assets, such as bills of exchange net of government bonds;
- interbank claims.

Interpretation: The asset categories allow the paper to observe how banks respond to funding stress: selling securities, reducing interbank lending, borrowing from other banks, or contracting credit.

2.4 Failure and distress classification

- The authors use Reichsbank records and historical sources to classify bank failures.
- Fifteen banks fail during or after the run in the main sample.
- Another seven banks receive government aid or are subject to distressed mergers.

- The main analysis contrasts failed banks with surviving banks, while robustness checks use broader definitions of distress.

Interpretation: The outcome is ex post bank failure, but the key explanatory variables are deposit flows observed during the run. This lets the authors test whether depositors reveal information before or during failure.

2.5 Stock-price data

- The authors hand-collect daily stock prices from the Berlin Stock Exchange's *Monatskursblatt*.
- Stock prices are available for 28 banks matched to balance sheet data.
- The stock-price sample is smaller than the deposit sample but allows a comparison with publicly available market information.

Interpretation: If stock prices identified failing banks as well as interbank flows, the interbank-market result might simply reflect public information. The paper finds weaker evidence from stock prices.

2.6 Historical timing of the 1931 run

The run has three broad phases:

1. **May 1931:** distress begins after the failure of the Austrian Creditanstalt. The interbank market starts contracting.
2. **June to early July 1931:** withdrawals intensify around reparations uncertainty, Nordwolle losses, and doubts about Germany's gold-standard commitment.
3. **July 10–14, 1931:** Reichsbank reserves fall below the legal coverage ratio; Danatbank and Dresdner Bank become illiquid; the government declares a bank holiday.

Interpretation: The timing matters because interbank withdrawals begin before the broad retail panic. Banks appear to act before regular depositors receive or process the same information.

3 Models and empirical specifications

3.1 Aggregate monthly dynamics

The paper estimates monthly bank-level dynamics using

$$y_{bt} = \gamma_b + \sum_{t \neq \text{Apr1931}} \beta_t \gamma_t + \varepsilon_{bt}, \tag{1}$$

where y_{bt} is the log of one plus a bank-level deposit or asset category, γ_b are bank fixed effects, and the coefficients are normalized to April 1931.

Interpretation: This specification tracks within-bank changes over time. The normalization makes April 1931 the pre-run benchmark.

3.2 Cross-sectional deposit-flow regression

The main cross-sectional regression is

$$\Delta y_{b,\text{Apr31:Jul31}} = \gamma_\theta + \beta_1 Failed_b + \beta_2 X_b + \varepsilon_b. \tag{2}$$

- $\Delta y_{b,\text{Apr31:Jul31}}$ is deposit growth from April to July 1931.
- $Failed_b$ equals one if bank b fails during or after the run.
- X_b includes leverage, liquidity, size quartiles, interbank funding, foreign funding, and Nordwolle exposure.
- γ_θ are bank-type fixed effects.

Interpretation: If regular depositors can identify weak banks, β_1 should be negative for regular deposits. If banks can identify weak banks, β_1 should be negative for interbank deposits.

3.3 Deposit dynamics for failing banks

To study timing, the paper estimates

$$y_{bt} = \gamma_b + \gamma_{\theta t} + \sum_{s \neq \text{Apr31}} \beta_s \mathbf{1}\{s = t\} Failed_b + \sum_{s \neq \text{Apr31}} \mu_s \mathbf{1}\{s = t\} X_b + \varepsilon_{bt}. \tag{3}$$

Interpretation: The coefficients β_s trace how failing banks differ from surviving banks month by month, controlling for bank characteristics and bank-type-time effects.

3.4 Failure-prediction regression

The predictive exercise reverses the outcome and estimates

$$Failed_b = \gamma_\theta + \beta_1 X_b + \beta_2 \Delta InterbankDeposits_b + \beta_3 \Delta Deposits_b + \varepsilon_b. \tag{4}$$

Interpretation: The model asks whether deposit flows during the run forecast which banks fail. Diagnostic performance is summarized by the area under the ROC curve, AUC.

3.5 Interbank-market functioning regression

The interbank exposure test uses

$$\begin{aligned}
\Delta InterbankExposure_{bt} = \gamma_b + \gamma_{\theta t} + \beta_1 \Delta RegDeposits/Assets_{bt} \quad (1)\\
+ \beta_2 \Delta RegDeposits/Assets_{bt} \times Post_t \quad (2)\\
+ \beta_3 \Delta RegDeposits/Assets_{bt} \times Failed_b \times Post_t \quad (3)\\
+ \beta_4 \Delta RegDeposits/Assets_{bt} \times Failed_b + \beta_5 Failed_b \times Post_t + \varepsilon_{bt}. \quad (5)
\end{aligned}$$

- $InterbankExposure$ equals interbank lending minus interbank borrowing.
- $Post_t$ turns on after April 1931.
- A functioning interbank market predicts $\beta_1 \approx 1$: banks with deposit inflows lend, banks with outflows borrow.
- A freeze for failing banks predicts a negative β_3.

Interpretation: This specification distinguishes a general interbank freeze from selective exclusion of failing banks. The evidence supports selective exclusion.

4 Identification and empirical design

4.1 What identifies depositor information

- All banks are exposed to the same systemwide run.
- Banks differ in ex post failure status and in observed deposit flows by depositor type.
- The main comparison is whether deposit outflows differ between failing and surviving banks within the same crisis window.
- Bank-type fixed effects absorb broad differences between Berlin banks, regional banks, Girozentralen, and Landesbanken.
- Balance sheet controls absorb observable fragility before the run.

Interpretation: The paper does not need a randomized shock to individual banks. The systemwide run creates a common stress test, and depositor flows reveal who could separate weak from strong banks.

4.2 Why the distinction between regular and interbank depositors matters

- Regular depositors withdraw broadly and similarly from failing and surviving banks.
- Interbank depositors withdraw selectively from failing banks.
- The same bank can be treated differently by different depositor types.

Interpretation: This is the central identification logic: if failure status were simply caused by total deposit outflows, regular deposits and interbank deposits should show similar patterns. They do not.

4.3 Why the paper argues banks are informed

- Failing and surviving banks lose similar total deposit funding.
- Interbank deposits are a small share of total funding, so interbank withdrawals alone are unlikely to be the primary mechanical cause of failure.
- Interbank flows predict failure even after controlling for observable balance sheet risk.
- The interbank market continues to lend to surviving banks with deposit outflows.

Interpretation: The evidence is most consistent with banks learning or knowing which counterparties are weak and withdrawing from them, rather than a universal funding-market freeze.

4.4 Why the interbank-market results are important

- In normal times, deposit inflow banks lend to deposit outflow banks.
- During the run, this risk-sharing relation remains for surviving banks.
- The relation breaks down for failing banks.

Interpretation: The interbank market is not merely a channel of contagion or hoarding. It is an information-sensitive liquidity allocation mechanism.

4.5 Limits of the design

- The setting is historical and Germany-specific.
- Savings banks and some private banks are not in the micro data.
- Failure is observed ex post, and the paper cannot perfectly separate information from causal funding effects.
- Stock-price evidence is limited by the small number of publicly traded banks.

Interpretation: The strongest claim is that interbank flows reveal superior information about bank failure risk. The paper is more cautious about proving that interbank outflows did not contribute at all to failure.

5 Tables and figures

5.1 Table 1 and Figures 1–2: balance sheet structure and aggregate run dynamics

Read:

- Table 1 shows that the average bank invests 73% of assets in illiquid assets and has 66% of liabilities in deposits.
- Regular deposits are the largest funding category, while domestic interbank deposits are important but smaller.
- Figure 1 shows a systemwide contraction in deposits and a large decline in interbank positions during the run.
- Figure 2 shows that the run starts in the interbank market: domestic interbank deposits fall around 10% in May and more than 30% by July.
- Regular deposits remain stable in May but fall by more than 15% by July; time deposits fall much more than demand deposits.

Economic message: The run begins as a run of banks on banks and only later becomes a broader depositor run.

Table 1
Bank assets and liabilities by share of total assets in spring 1931.

A. Assets

Bank type	Illiquid assets			Liquid assets			Interbank		Assets (in mil. RM)	No. of banks
		Loans	Covered bonds		High	Low	Total	Short-term		
All banks	0.73	0.52	0.15	0.15	0.05	0.10	0.09	0.04	214	126
Berlin banks	0.64	0.60	0.00	0.21	0.06	0.15	0.14	0.07	2,088	6
Girozentralen	0.74	0.21	0.49	0.08	0.06	0.02	0.17	0.03	300	17
Regional banks	0.74	0.63	0.03	0.16	0.04	0.12	0.07	0.04	52	82
Landesbanken	0.75	0.32	0.39	0.12	0.07	0.06	0.11	0.03	241	21

B. Liabilities

Bank type	Deposits							Acceptances	Bonds	Equity
		Demand	Time	Regular	Domestic bank	Other	Foreign			
All banks	0.66	0.22	0.40	0.51	0.23	0.03	0.10	0.03	0.14	0.17
Berlin banks	0.86	0.27	0.42	0.60	0.11	0.17	0.77	0.05	0.01	0.07
Girozentralen	0.49	0.20	0.29	0.17	0.69	0.00	0.00	0.02	0.44	0.05
Regional banks	0.70	0.23	0.44	0.61	0.10	0.03	0.07	0.03	0.03	0.23
Landesbanken	0.56	0.19	0.36	0.36	0.39	0.01	0.10	0.02	0.38	0.05

Table 1: Bank assets and liabilities in spring 1931

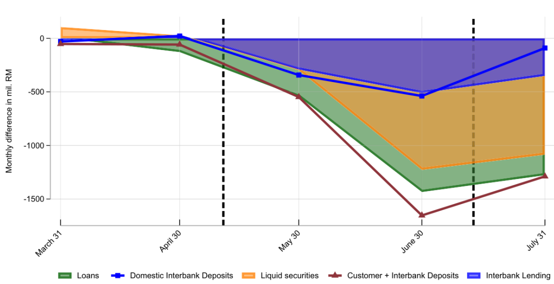


Figure 1
Aggregate dynamics of assets and liabilities
The figure shows the month on month aggregate changes of key balance sheet components during the crisis in

(a) Figure 1: aggregate asset/liability dynamics

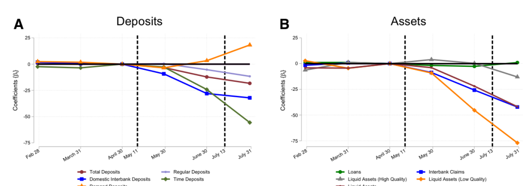


Figure 2
Deposit and asset dynamics during Spring 1931

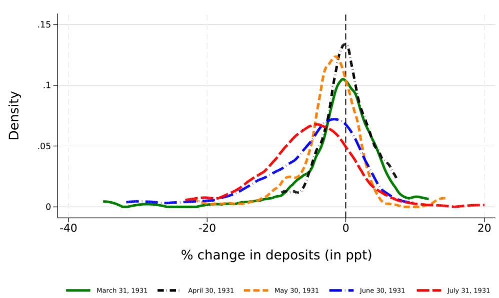
(b) Figure 2: monthly deposit and asset dynamics

5.2 Figure 3, Table 2, and Figure 4: do failing banks lose more deposits?

Read:

- Figure 3 shows substantial cross-sectional heterogeneity in deposit flows. Even during the run, some banks receive inflows.
- Table 2 shows no significant difference in regular-deposit outflows between failing and surviving banks.
- The key coefficient for interbank deposits is very large: failing banks lose about 70–76 percentage points more interbank deposits than surviving banks.
- Total deposit flows do not differ significantly between failing and surviving banks.
- Figure 4 visualizes the same point: regular-deposit distributions overlap, while interbank-deposit distributions differ sharply.

Economic message: Regular depositors do not identify failing banks. Interbank depositors do.



(a) Figure 3: cross-section of deposit flows

Table 2
Deposit flows from April 1931 through July 1931 for both failing and surviving banks.

Dep. variable	Regular		Interbank		Demand		Time		Total	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Failed	−2.1	−4.3	−69.7***	−76.1***	−4.9	−2.8	−12.0	−15.3*	−0.8	−2.5
	(4.4)	(4.4)	(24.5)	(25.0)	(12.3)	(11.5)	(8.6)	(7.8)	(3.8)	(3.5)
Leverage		−0.1		0.7		1.0**		−0.3		0.0
		(0.2)		(0.9)		(0.4)		(0.3)		(0.1)
Liquidity		−21.0***		−22.1		−4.3		−23.4**		−15.7***
		(5.6)		(32.0)		(14.7)		(9.9)		(4.5)
2nd Size quartile		−3.1		−10.9		−7.1		−12.9**		−3.8
		(3.5)		(19.9)		(9.2)		(6.2)		(2.8)
3rd Size quartile		2.9		17.3		17.6		−5.9		5.0
		(4.0)		(22.9)		(10.6)		(7.1)		(3.2)
4th Size quartile		5.1		42.4		24.7		−7.2		5.0
		(5.7)		(32.3)		(14.9)		(10.0)		(4.6)
Interbank Funding		0.9		20.3		64.8***		−64.4***		−8.9
		(8.8)		(50.2)		(23.2)		(15.6)		(7.1)
Foreign Funding		−6.2		−72.4***		−8.6		−11.6		−11.3***
		(4.7)		(26.4)		(12.2)		(8.2)		(3.7)
Nordwolle Connection		−5.4		18.7		6.5		−14.3		−5.0
		(7.4)		(42.0)		(19.4)		(13.0)		(6.0)
Number of Banks	118	118	118	118	118	118	118	118	118	118
Bank Type FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
R^2	.0019	.15	.067	.15	.0014	.24	.017	.3	.00045	.23

This table reports results from estimating

(b) Table 2: deposit flows, failed vs. surviving banks

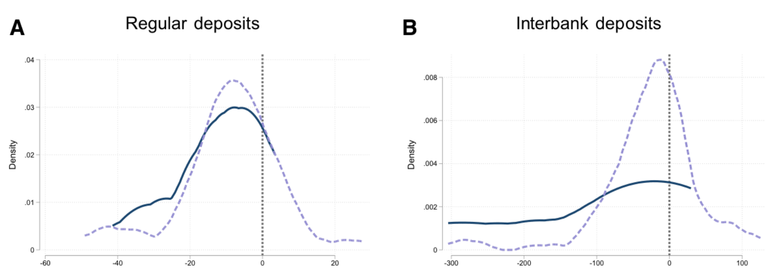


Figure 4
Deposit growth from April 1931 through July 1931 for failing and surviving banks

Figure 4: density of regular and interbank deposit growth

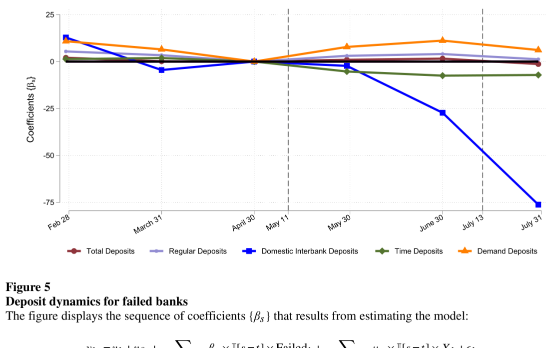
5.3 Figure 5, Table 3, and Figures 6–7: predicting which banks fail

Read:

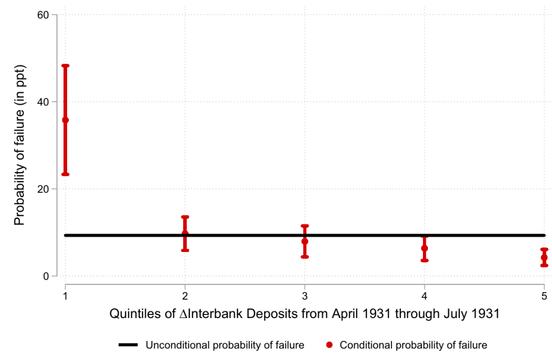
- Figure 5 shows that failing banks lose access to the interbank market over time; domestic interbank deposits fall by roughly 75 percentage points by July.

- Table 3 shows that interbank deposit growth predicts failure. Observable controls alone produce an AUC near 0.58, while adding interbank deposit flows raises AUC above 0.8 in the strongest specifications.
- Figure 6 shows that banks in the worst quintile of interbank deposit growth have a much higher failure probability than banks in other quintiles.
- Figure 7 shows that the nonlinear interbank-deposit model has the strongest ROC curve.

Economic message: Interbank withdrawals are not just noise; they contain information about which banks will fail.



(a) Figure 5: deposit dynamics for failed banks



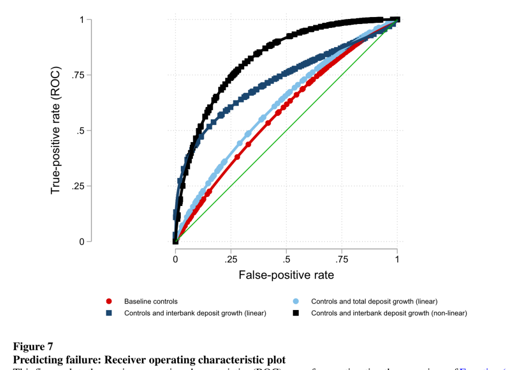
(b) Figure 6: failure probability by interbank-flow quintile

Table 3
Predicting bank failure with and without deposit flows.

Dependent variable	Failed							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ΔInterbank Deposits$_{\text{April 31:July 31}}$			-11.57^{**}		-11.10^{**}		-11.79^{*}	-11.46^{**}
			(5.51)		(4.84)		(6.19)	(5.45)
ΔDeposits$_{\text{April 31:July 31}}$				−30.00		−25.32	4.24	6.84
				(27.16)		(25.88)	(32.48)	(30.43)
Leverage	0.01	0.17	−0.02	−0.01	0.23	0.18	−0.02	0.23
	(0.12)	(0.18)	(0.12)	(0.12)	(0.18)	(0.18)	(0.11)	(0.18)
2nd Size quartile	11.04	−2.41	8.35	9.40	−3.58	−3.56	8.52	−3.31
	(8.65)	(9.50)	(8.26)	(8.83)	(8.90)	(9.71)	(8.17)	(8.90)
3rd Size quartile	−1.36	−12.37	0.41	−0.05	−9.49	−11.14	0.26	−9.73
	(6.75)	(8.67)	(6.86)	(6.63)	(8.07)	(8.49)	(6.84)	(8.06)
4th Size quartile	−4.90	−6.03	−0.31	−3.46	−1.04	−5.00	−0.43	−1.15
	(6.57)	(8.61)	(6.33)	(6.37)	(8.34)	(8.60)	(6.17)	(8.31)
Foreign Funding	0.95	−2.48	−7.34	−3.02	-10.55^{*}	−5.61	−6.94	−9.96
	(3.52)	(4.94)	(5.34)	(3.89)	(6.34)	(5.36)	(4.71)	(6.02)
Liquidity	6.14	−15.30	1.69	0.43	−16.78	−19.58	2.41	−15.67
	(11.57)	(11.08)	(10.68)	(13.09)	(11.01)	(12.55)	(13.25)	(13.10)
Interbank	4.38	22.72	2.22	2.02	23.02	20.28	2.52	23.69
	(8.80)	(19.02)	(9.06)	(8.19)	(19.11)	(18.43)	(8.33)	(18.70)
Nordwolle Connection	17.61	20.13	19.06	15.02	19.64	17.68	19.46	20.29
	(15.32)	(16.93)	(15.31)	(13.88)	(16.20)	(15.38)	(14.78)	(15.73)
Number of Banks	118	118	118	118	118	118	118	118
Bank-Type FE	No	Yes	No	No	Yes	Yes	No	Yes
R^2	.073	.17	.15	.083	.24	.17	.15	.24

main

(a) Table 3: predicting bank failure



(b) Figure 7: ROC curves

5.4 Table 4, Figure 8, and Table 5: does the interbank market freeze?

Read:

- Table 4 restricts the sample to banks with relatively little reliance on interbank funding before the run.
- Even for these banks, regular and total deposit flows are similar across failing and surviving banks, but interbank flows are sharply lower for failing banks.
- Figure 8 shows that before the crisis, banks with regular-deposit inflows increase interbank exposure and banks with outflows reduce it.
- During the run, surviving banks maintain this relation; failing banks do not.

- Table 5 confirms this regression result: the baseline slope is about 0.90, and the triple interaction for failing banks during the run is around −0.5.

Economic message: The interbank market does not collapse for everyone. It continues to provide liquidity to surviving banks but excludes banks that will fail.

Table 4
Deposit flows from April 1931 through July 1931 for failed banks by reliance on interbank market funding as share of total deposits funding.

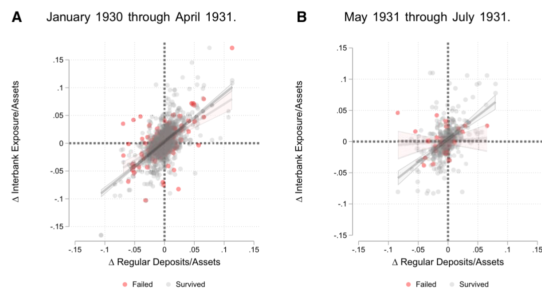
Dep. variable	Regular			Interbank			Total		
Interbank share (in %)	< 10%	< 7.5%	< 5%	< 10%	< 7.5%	< 5%	< 10%	< 7.5%	< 5%
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Failed	−0.5	−0.2	1.9	−135.5***	−124.7***	−112.9**	−2.2	0.8	3.1
	(4.4)	(5.1)	(4.8)	(36.3)	(41.6)	(44.6)	(4.3)	(4.7)	(4.6)
Mean LHS	−8.9	−7.8	−9	−41	−52	−53	−11	−11	−11
Number of Banks	66	49	40	66	49	40	66	49	40
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bank Type FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2	.33	.22	.38	.36	.46	.43	.43	.47	.48

This table reports results from estimating

$$\Delta y_{b\text{April 31:July 31}} = \gamma_\theta + \beta_1 \times \text{Failed}_b + \beta_2 \times X_b + \epsilon_b,$$

where $\Delta y_{b\text{April 31:July 31}}$ is the log-growth in the type of deposit indicated in the table header. We restrict the sample to banks that use less than 10%, 7.5%, or 5% of interbank funding as a share of total deposit funding as indicated in the table header. As control variables, we include a bank's ratio of total liabilities (total assets net of equity) to equity, liquid assets (securities and interbank claims) to total deposits, interbank funding to total deposits, indicators of the size quartile based on total assets, an indicator for use of foreign-currency denominated

(a) Table 4: low-interbank-reliance subsamples



(b) Figure 8: interbank exposure and regular-deposit flows

Table 5
Interbank exposure and regular deposit flows before and during the run.

Dependent variable	Δ Interbank Exposure		
	(1)	(2)	(3)
ΔReg. Deposits/Assets	0.90***	0.90***	0.85***
	(0.06)	(0.06)	(0.08)
ΔReg. Deposits/Assets×Post	−0.15	−0.10	−0.14
	(0.09)	(0.09)	(0.13)
ΔReg. Deposits/Assets×Post×Failed	−0.51**	−0.54**	−0.49*
	(0.22)	(0.24)	(0.25)
ΔReg. Deposits/Assets×Failed	−0.20	−0.17	−0.12
	(0.16)	(0.17)	(0.17)
Post×Failed	−0.00	−0.00	−0.00
	(0.00)	(0.00)	(0.00)
Bank Sample	All	All	Berlin + Regional
N	1732	1732	1214
Number of Banks	130	130	92
Time FE	Yes	No	Yes
Bank Type Time FE	No	Yes	No
R^2	0.4	0.42	0.39

This table reports results from estimating:

$$\Delta\text{Interbank Exposure}_{bt} = \gamma_b + \gamma_{\theta t} + \beta_1 \times \Delta\text{Reg. Deposits/Assets}_{bt}$$

$$+ \beta_2 \times \Delta\text{Reg. Deposits/Assets}_{bt} \times \text{Post}_t$$

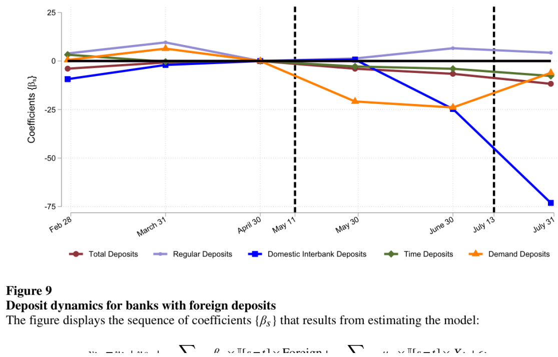
Table 5: interbank exposure and regular deposit flows

5.5 Figure 9, Figure 10, and Table 6: foreign deposits and stock prices

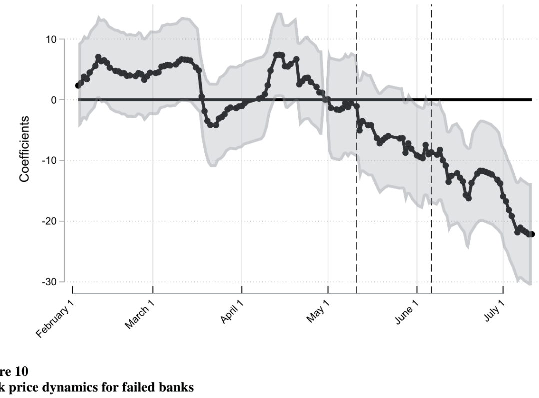
Read:

- Figure 9 shows that banks exposed to foreign-currency deposits lose more domestic interbank deposits.
- This supports the idea that banks had information about hidden vulnerabilities, because foreign-currency exposure was not publicly visible.
- Figure 10 shows that stock prices of failing banks decline during the run, especially after May.
- Table 6 shows little systematic relation between interbank deposit flows and risk-adjusted bank stock returns.
- The stock market evidence is weaker than the interbank-flow evidence, partly because only 28 banks have matched stock prices.

Economic message: Interbank flows appear to reveal information that is not fully summarized by public stock prices.



(a) Figure 9: banks with foreign deposits



(b) Figure 10: stock-price dynamics for failed banks

Table 6
Interbank Deposit Flows and Bank Stock Return.

Dependent variable	Average risk-adjusted daily returns				
Sample period	April - July	April 1931	May 1931	June 1931	July 1931
	A: Interbank deposit flows prior month				
	(1)	(2)	(3)	(4)	(5)
Prior Month Interbank	−0.028	−0.139	−0.016	0.088*	−0.129
	(0.062)	(0.279)	(0.078)	(0.046)	(0.169)
N	1635	475	471	477	212
No of Banks	28	25	27	24	24
R^2	.00011	.00099	.000072	.0016	.0038
	B: Interbank deposit flows current month				
	(1)	(2)	(3)	(4)	(5)
Interbank	−0.015	0.006	−0.020	−0.058	0.013
	(0.042)	(0.105)	(0.073)	(0.059)	(0.076)
N	1566	475	441	456	194
No of Banks	26	25	25	23	22
R^2	.000039	3.5e-06	.00011	.0007	.000062

Table 6: interbank flows and bank stock returns

6 Short summary

- The paper studies whether depositors can tell which banks will fail during the German banking crisis of 1931.
- The setting has no deposit insurance and little regulation, so depositors had incentives to withdraw from weak banks.
- Newly digitized bank balance sheets allow monthly tracking of different deposit categories across more than 120 banks.
- Aggregate deposits fall by roughly 20% during the run.
- Total deposit outflows are similar for failing and surviving banks.
- Regular depositors do not distinguish failing from surviving banks.
- Interbank depositors do: interbank deposits collapse almost exclusively at failing banks.
- Interbank deposit outflows predict future failure, with AUC values above 0.8 in the best specifications.
- The interbank market remains active for surviving banks with deposit outflows.
- Banks with inflows do not simply hoard liquidity; they lend to other surviving banks.
- Failing banks are excluded from interbank funding.
- Banks with foreign-currency deposit exposure lose more interbank funding, suggesting banks detected hidden vulnerabilities.
- Stock prices of failing banks decline, but stock-market evidence is weaker than interbank-flow evidence.
- The main lesson is that banks are more informed than regular depositors and selectively withdraw from weak counterparties.

7 Conclusion

7.1 Core conclusion

Blickle, Brunnermeier, and Luck show that during the German banking crisis of 1931, not all depositors were equally informed. Regular depositors ran broadly and did not separate failing from surviving banks. Banks, acting as interbank depositors, withdrew selectively from banks that would fail.

7.2 Economic intuition

A bank has incentives and ability to monitor other banks because its exposure is large, repeated, and professionally managed. A retail or nonfinancial depositor faces higher information costs and often reacts later to public failure events. The result is a two-layer run: first an informed interbank run against weak banks, then a broader depositor run once the crisis becomes public.

7.3 Takeaway

The paper revises the simple view that interbank markets always freeze in crises. In 1931 Germany, the interbank market collapsed for failing banks but continued to allocate liquidity to surviving banks. Interbank flows therefore served as both an information signal and a liquidity-allocation mechanism.